

Fractional-Width ResNets and the Spectral Anatomy of Double Descent Localizing the Phase Transition to $k \in [0.125, 0.25]$ on Noisy CIFAR-10

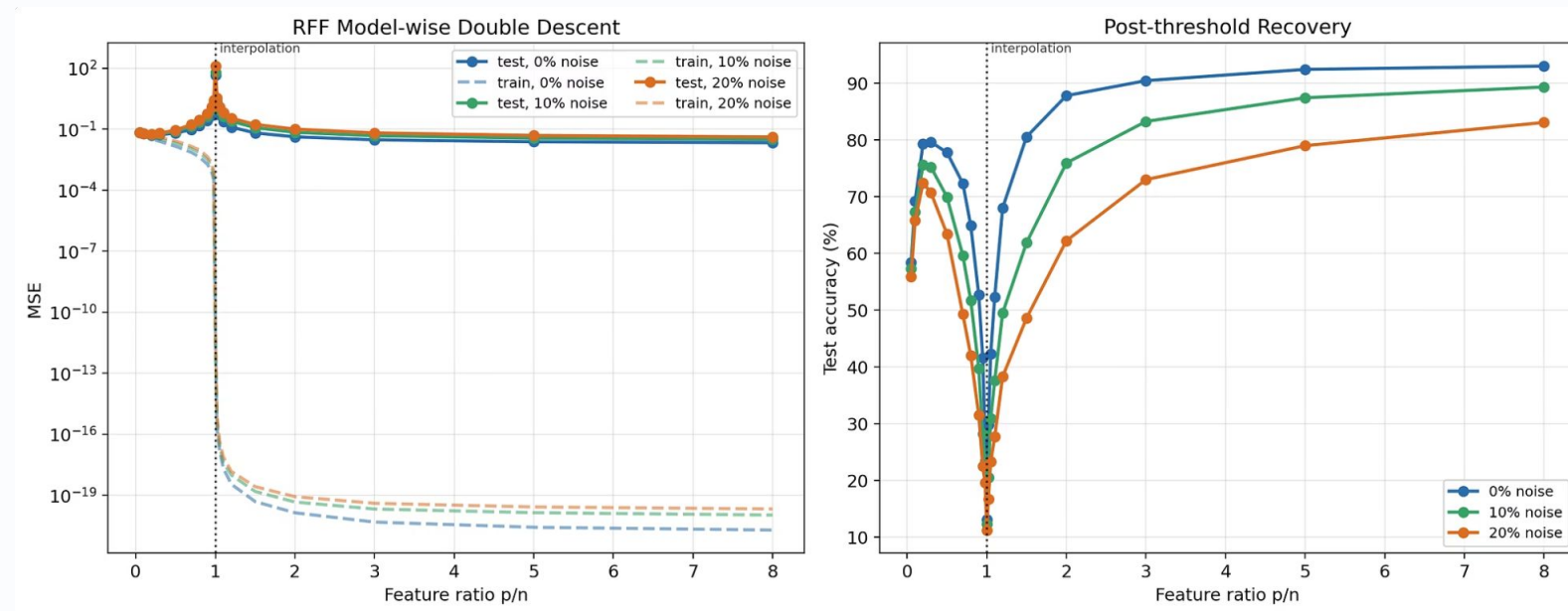
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What Did We Learn About a Neural Network?

- Three themes: approximation, optimization landscape, generalization
 - Two scaling regimes: NTK / lazy training vs. mean-field / active training
 - Our angle: where do over-parametrized ResNets sit on the DD curve?
-
- Width is the right axis to span the interpolation threshold $k \in [0.125, 0.25]$

From Bias-Variance to Double Descent



RFF model-wise double descent on MNIST — test error spikes at $p/n = 1$, then recovers. Belkin et al. (2019).

- Classical bias-variance (L1–L3): more parameters $\rightarrow$ more variance $\rightarrow$ test error rises past $p \approx n$. U-shaped in p .
- Belkin (2019) / Nakkiran (2021): error rises near interpolation, then descends
- EMC (Nakkiran §3): peak occurs at $\text{EMC} \approx n$, not literal $p \approx n$
- Three DD axes: model-wise (sweep p), sample-wise (sweep n), epoch-wise (sweep t)



$$\text{EMC}(T) := \max\{n : E[\text{train error}] = 0\}$$

Why Literal ResNet-18 Can't Show DD — and What We Built

The Problem

- torchvision ResNet-18 is 4-stage with base widths (64, 128, 256, 512)
- Width-multiplier sweep at $n = 4,000$, 15% noise:

width ×	params	train acc	test acc
0.5	2.80 M	100.0%	54.08%
1.0	11.17 M	99.9%	56.62%
2.0	44.66 M	100.0%	60.94%

- All three at $p/n \in [700, 11,200]$ — already deep in the over-param tail
- All three memorize → discrete BasicBlock widths can't cross the interpolation threshold

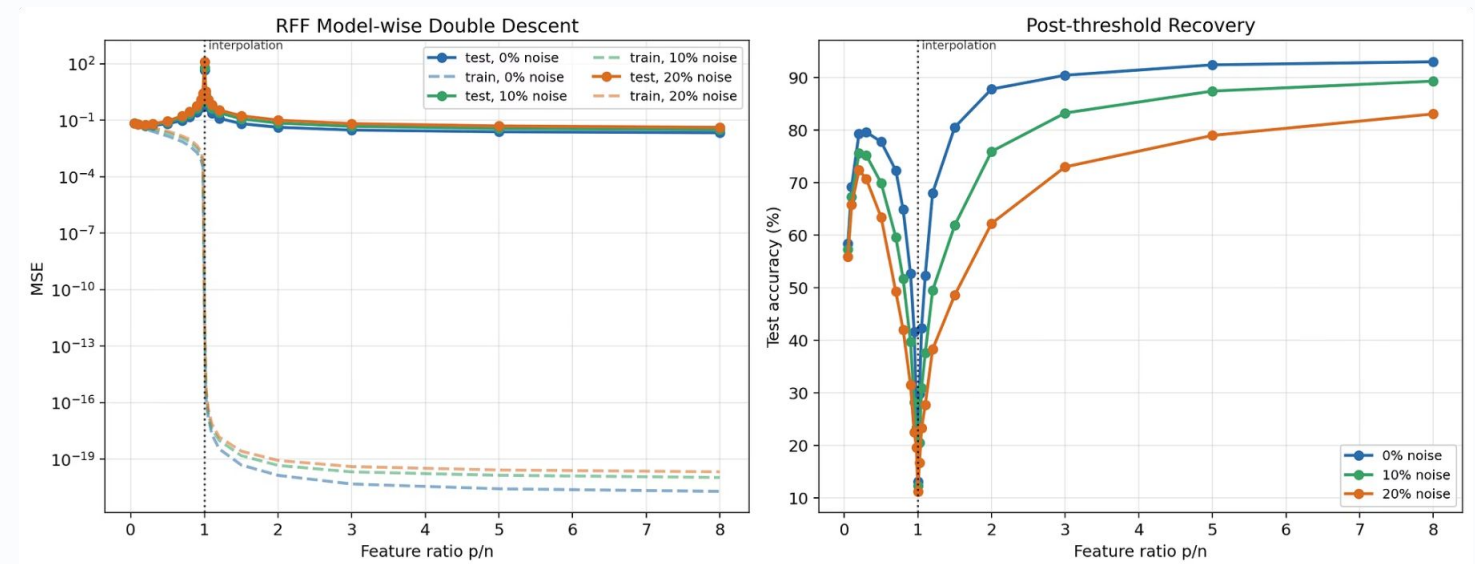
What We Built — ResNetK

- A 3-stage fractional-width ResNet with continuous capacity
- Auto-switch BatchNorm → GroupNorm when $\min(c_i) \leq 2$
- $k \in \{0.0625, 0.125, 0.1875, 0.25, 0.375, 0.5, 0.75, 1.0, 2.0\}$
- 9 points span $p \in [823, 696,618]$ — an 850× range
- At $k = 0.0625$: widths (1, 2, 4), train acc 25% ⇒ genuinely under-parameterized
- Same trained checkpoints reused for spectral diagnostics

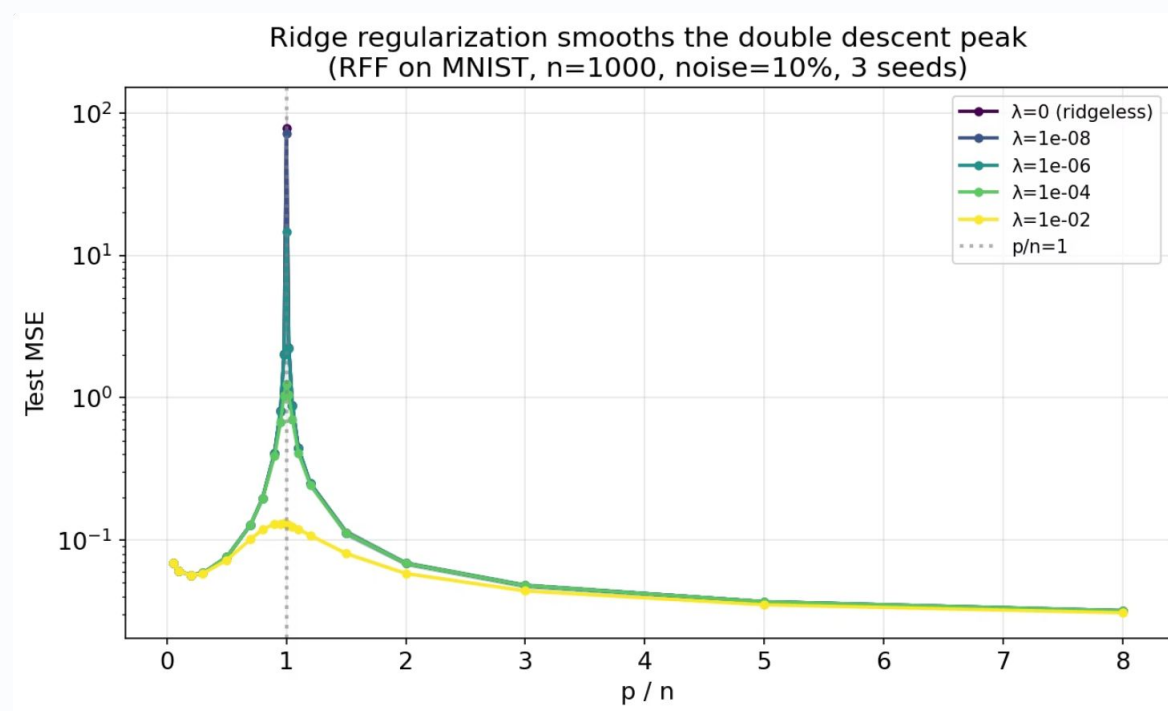
Identical hyperparameters across all k : Adam lr $1e-4$ · 2,000 epochs · 15% label noise · 2 seeds (headline curve). Spectral & Hessian witnesses computed at seed 42 on the same trained checkpoints.

Reproduction — RFF Model-Wise Double Descent

- Random Fourier features: $\varphi(x) = \sqrt{2/D} \cos(Wx + b)$,
 $D = p$
- Test error spikes near $p/n = 1$, then descends as p grows past n
- Classical U-curve only holds restricted to $p < n$
- Replicates Belkin et al. (2019) Figure 1 on MNIST



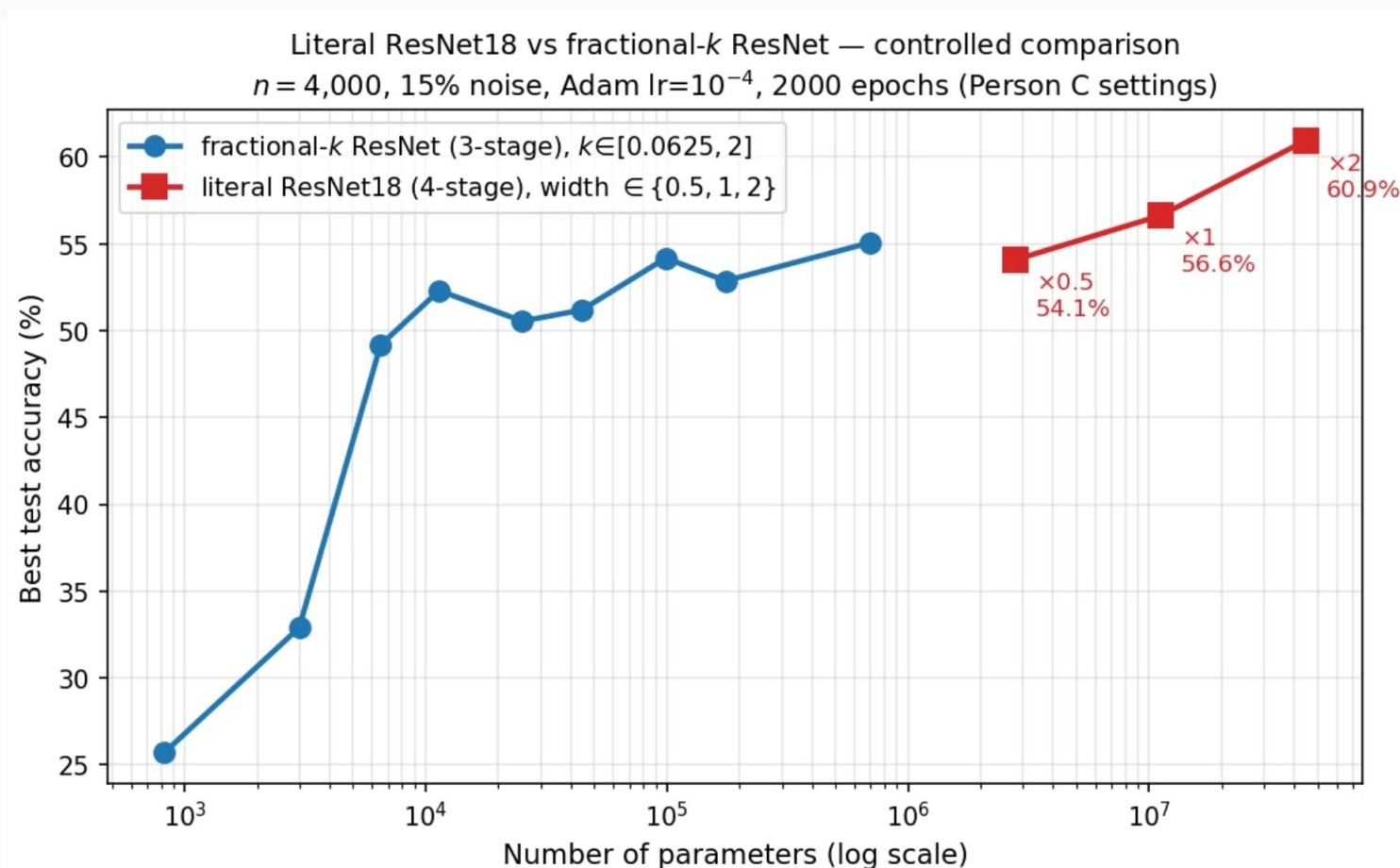
Reproduction – RFF Mechanism: Variance and Ridge Smoothing



Ridge sweep $\lambda \in [10^{-6}, 10^{-2}]$ on RFF; peak smooths monotonically

- Gram matrix $\Phi^T \Phi$ has condition $\kappa \rightarrow \infty$ at $p/n = 1$
 - MMin-norm interpolant amplifies label noise at the threshold
 - Ridge $\lambda > 0$ shrinks small singular values; smooths the peak monotonically
 - $\lambda \in [10^{-6}, 10^{-2}]$ sweep; by $\lambda = 10^{-2}$ the U-shape vanishes (L10)
- Noisexridge extension: λ suppresses 0–40% noise-amplified peaks by reducing norm, κ , and effective df

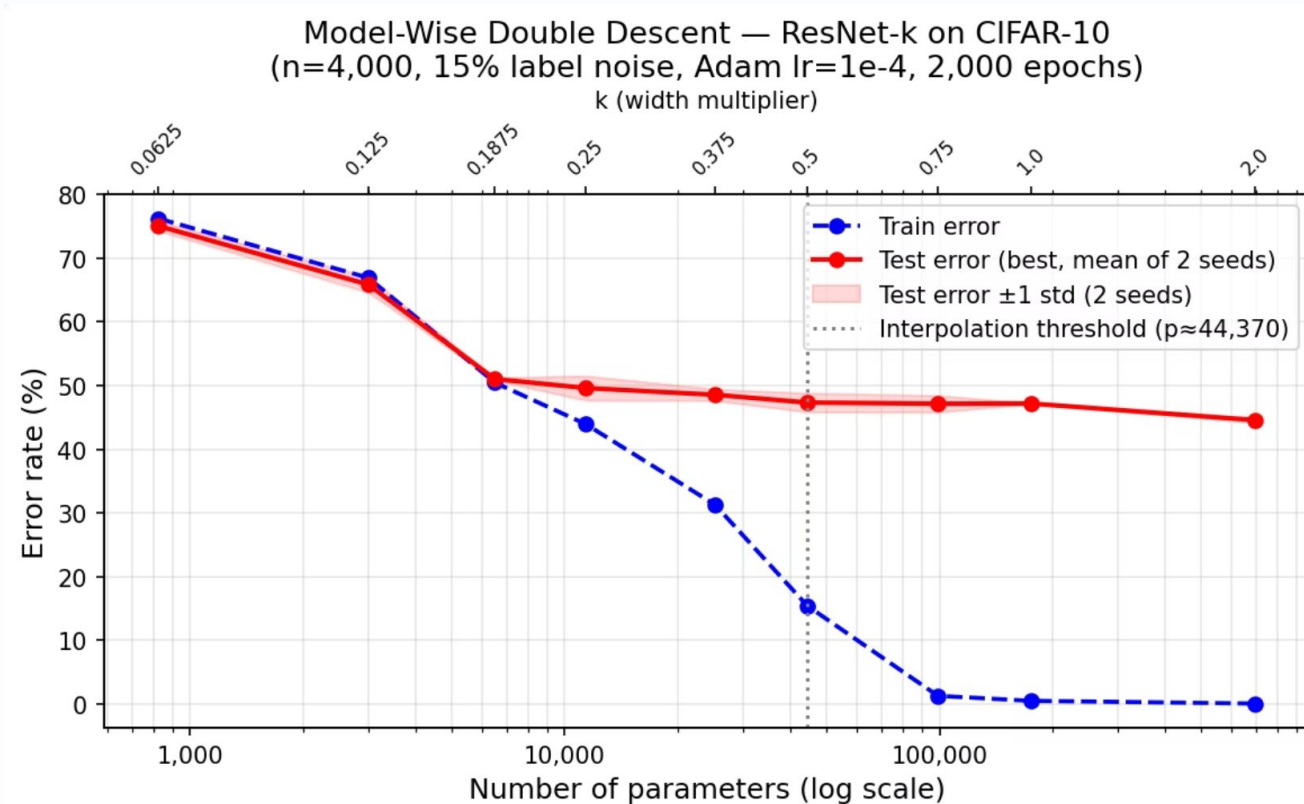
Puzzle – Literal ResNet-18 Fails To Reveal DD



Literal ResNet-18 (red) vs fractional-k ResNet (blue) — controlled comparison; $n = 4000$, 15% noise, Adam $\text{lr} = 10^{-4}$, 2000 epochs

- torchvision ResNet-18, width $\times \{0.5, 1, 2\}$; $n = 4000$, 15% noise, identical hparams
- Test accuracy 54.08% / 56.62% / 60.94% — monotonically increasing
- All three at $p/n \in [700, 11200]$; all memorize ($\sim 100\%$ train acc)
- Discrete BasicBlock widths cannot cross the interpolation threshold
- **Conclusion: standard ResNet-18 skips the threshold; fractional width is needed to reveal DD.**

Fractional-Width ResNet Recovers Double Descent



Model-wise risk vs. width: train error and test error (100 – test accuracy) from best checkpoint. Top axis: width multiplier k ; bottom axis: parameter count

- $k = 0.0625 \rightarrow 24.9\%$ (underfitting; high error on the left)
- $k = 0.1875 \rightarrow 49.0\%$ (transition: test error stops improving “for free”)
- $k = 2.0 \rightarrow 55.4\%$ (over-param tail: further gain vs. mid-k plateau)

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- Rise (small k) $\rightarrow$ transition $\rightarrow$ flat mid-k
- Not a sharp DD spike; mid-k “U” is shallow (~ 1 pp)

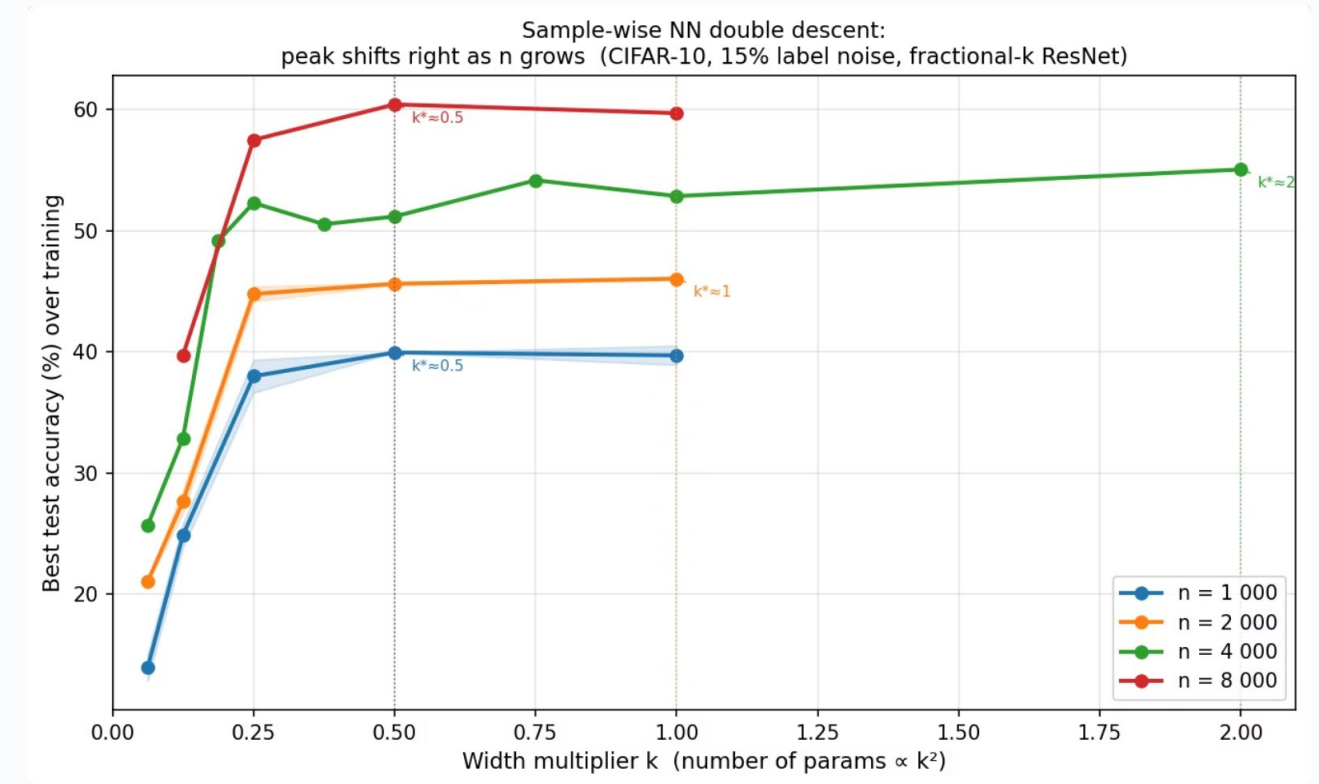
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Sample-Wise DD — $k^*(n)$ Migrates with n

- For $n = 1k \rightarrow 2k \rightarrow 4k$, k^* moves right ($\approx 0.5 \rightarrow 1.0 \rightarrow 2.0$ in our runs)
- Qualitatively matches EMC-style “matching complexity rises with n ” — direction only, not a theorem.
- Coarse / uneven k -grids $\rightarrow$ trend statement only; no scaling-law fit; cautious comparing k^* across all n .

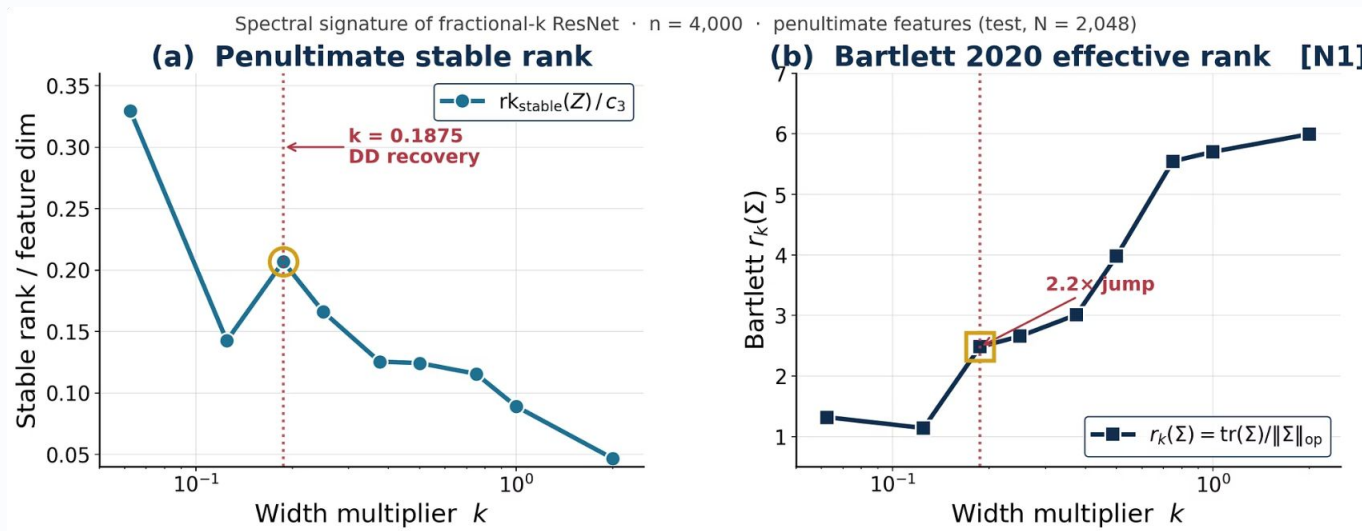


- Trend: more data favors larger k on our sweep.
- k^* is a grid argmax, not a continuous optimum.



Best test acc. (%) — mean over seeds; k^* is the k with highest mean acc on the sweep shown. Coarser k -sweeps for some n : compare k^* only as a trend, not as exact optima

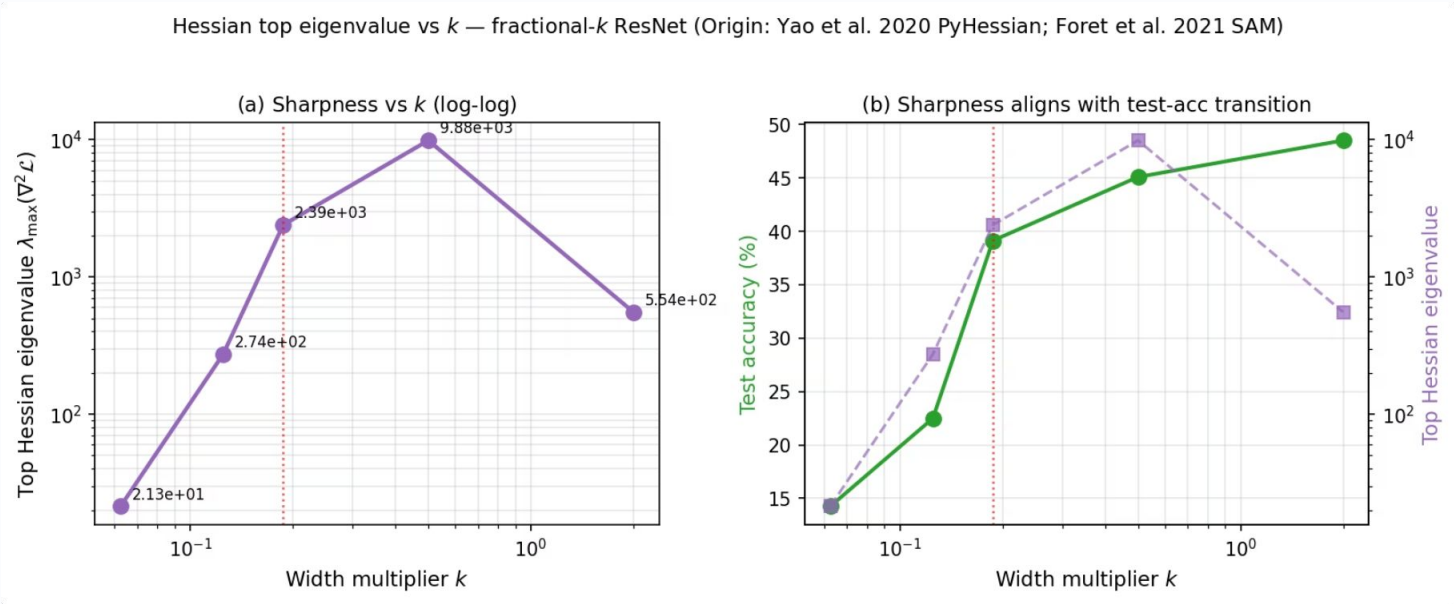
Mechanism I – Four Spectral Witnesses



Penultimate spectrum vs. k ($n = 4,000$, test $N = 2048$); normalised by c_3 ; line $k = 0.1875$ matches DD recovery width.

- Stable rank is non-monotonic in k , with a structural feature near $k \approx 0.1875$
- Penultimate-layer condition number κ spikes in the same regime, indicating local ill-conditioning
- Participation ratio exhibits a matching inflection, suggesting a shift in effective dimensionality
- Transition band $k \in [0.125, 0.25]$, consistent with **model-level recovery**

Mechanism II – Hessian Sharpness, 5th Witness

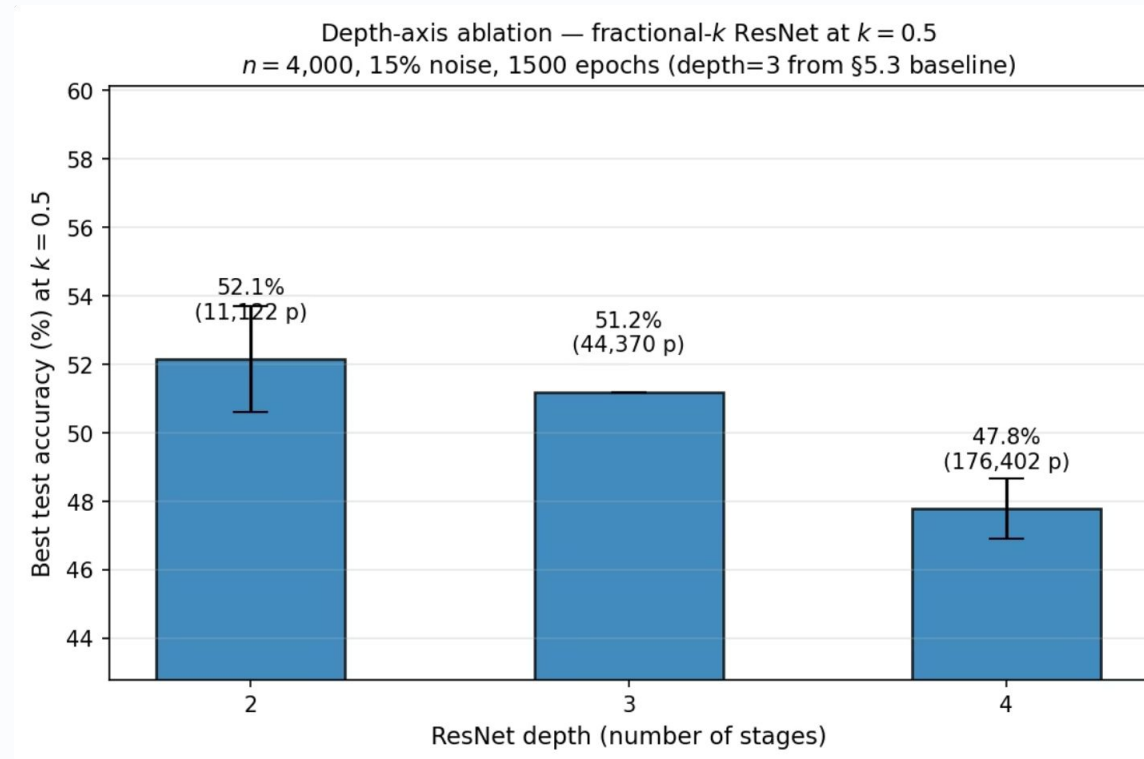


$\lambda_{\max}(\nabla^2 \mathcal{L})$ vs k ; absolute peak at $k = 0.5$, 18× drop at $k = 2$ (SAM regime);
per-parameter version peaks at $k = 0.1875$.

Yao 2020 PyHessian · Foret 2021 SAM

- Method: $\lambda_{\max}(\nabla^2 \mathcal{L}(\theta))$ via 30-step power iteration on 256 samples (Yao 2020 PyHessian).
- Absolute peak at $k = 0.5$; 18× drop at $k = 2$ — the SAM flat-minimum signature (Foret 2021).
- Per-parameter $\lambda_{\max} / p$ peaks at $k = 0.1875$ — same width as the four spectral witnesses.**
- Two phase boundaries together: spectral collapse near recovery onset, landscape flattening on the way to dense.
- Full trajectory in caption; figure shows the curve, not the table.

Robustness — Depth Ablation



Depth ablation at $k = 0.5$; $n = 4000$, 15% noise, 1500 epochs.

- Depth ablation at $k = 0.5$: depth 2 $\rightarrow$ 52.14%, depth 3 $\rightarrow$ 51.18%, depth 4 $\rightarrow$ 47.77%.
- DD-recovery shape preserved across depth; lean depth wins on noisy data.
- Optimizer is secondary, not primary: Adam and SGD both fail at $p/n \geq 2.8$ — test accuracy collapses to 5–7%. EMC and width are the primary axes.**
- Width axis dominates; depth and optimizer are supporting controls.

Takeaways

- Parameter count is the wrong axis: 55.4% test accuracy at $p/n = 174$ ($k = 2$).
- One mechanism, five witnesses, all concentrating at $k \in [0.125, 0.25]$.
- Recovery is activation-agnostic — GELU 49.88%, ReLU 48.19%, Tanh 46.15% at $k = 0.1875$. The Bartlett effective rank is a useful diagnostic, not a tight bound: vacuity ratio 1.0 at small width climbs to 3.4 at $k = 2$.**
- Width is the cleanest axis to span the interpolation threshold; literal ResNet-18 is monotone.
- Two phase boundaries: spectral collapse at recovery onset, landscape flattening from $k = 0.5$ to $k = 2$.
- Reproducibility: `make figures` regenerates every plot from results JSONs.

L1–L3 classical bias-variance · L4–L6 Hessian · L7–L8 NTK · L9 Bartlett · L10 ridge mechanism · L11–L12 DD curve

Thanks: Yizheng for §6.11 epoch-wise; Z. Li for full empirical NTK; Y. Li for the make-figures pipeline and the activation ablation.

Thank You, any questions?

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Repo: github.com/shufengc/double-descent-deep-learning